

■ 2.5.9 Debugging

Whenever you write long programs in *Mathematica*, or for that matter any other language, it is inevitable that you will make mistakes in them from time to time. You will often want to execute your program in steps, so that you can see exactly what it is doing.

`Debug[expr]` allows you to step through the evaluation of *expr*. Exactly how `Debug` works will depend on the computer system you are using.

<code>Debug[expr]</code>	step through the evaluation of <i>expr</i>
<code>Debug[expr, {f, g, ...}]</code>	step through the evaluation of all calls to the functions <i>f</i> , <i>g</i> , ...
<code>On[f]</code>	display a trace of all calls to the function <i>f</i>
<code>Off[f]</code>	stop tracing <i>f</i>
<code>On[]</code>	trace of all function evaluations
<code>Off[]</code>	stop all tracing

Functions for tracing and debugging.

When you stop the execution of a program in `Debug`, you can “inspect” the state you are in. You can do any operation, so long as it does not change the state. You cannot use any *Mathematica* function that resets the value of any variable.

<code>i</code>	inspect current state
<code>s</code>	step through execution
<code>n</code>	continue to next expression
<code>c</code>	continue to the end of the current block
<code>f</code>	finish computation
<code>a</code>	abort computation

Typical options when you stop the execution of a program in `Debug`.